

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 05

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Recap of Previous Lecture



Topic

Stop-and-Wait Protocol ✓





Topics to be Covered



Topic

Sliding Window Protocol



ABOUT ME



Hello, I'm **Abhishek**

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[MCQ]

[GATE-2025] [2 Mark]



#Q. Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?

- ☒ A 88.23
- ☐ B 93.75
- ☐ C 85.44
- ☐ D 66.67

Ans: A

Solution:-

$$\text{Frame Size} = 3000 \text{ bits}$$

$$\text{Bandwidth} = 2000 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{3000 \text{ bits}}{2000 \text{ bits / sec}} = 1.5 \text{ Sec}$$

$$t_p = 100 \text{ ms} = 0.1 \text{ Sec}$$

$$\text{Cycle time} = (t_x + 2 * t_p) = 1.7 \text{ Sec}$$

For Stop-and-Wait ARQ :

$$\text{Efficiency } (\eta) = \frac{\text{Transmission delay}}{\text{Cycle Time}} = \frac{1.5 \text{ Sec}}{1.7 \text{ Sec}}$$

$$= 0.8823$$

$$= 0.8823 * 100 \%$$

$$= 88.23 \%$$

[NAT]

[GATE-2017] [2 Mark]



#Q. The values of parameters for the Stop-and-Wait ARQ protocol are as given below.

Bit rate of the transmission channel = 1Mbps

Propagation delay from sender to receiver = 0.75 ms

Time to process a frame = 0.25ms

Number of bytes in the information frame = 1980

Number of bytes in the acknowledge frame = 20

Number of overhead bytes in the information frame = 20 *

Assume that there are no transmission errors. Then the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal place).

IIT-R Ans: 86.5 to 89.5

Solution:-

$$\text{Packet Size} = 1980 \text{ bytes} \quad (\text{including overhead})$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{1980 \text{ bytes}}{10^6 \text{ bits / sec}} = 15.84 \text{ ms}$$

$$\text{Packet Size} = (20 + 1980) \text{ bytes} = 2000 \text{ bytes} = 16 * 10^3 \text{ bits}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{16 * 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 16 \text{ ms}$$

$$\text{ACK Size} = 20 \text{ bytes} = 160 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$t_{xA} = \frac{\text{ACK Size}}{\text{Bandwidth}} = \frac{160 \text{ bits}}{10^6 \text{ bits / sec}} = 0.16 \text{ ms}$$

$$t_p = 0.75 \text{ ms}$$

$$\text{Processing Delay} = 0.25 \text{ ms}$$

$$\underline{t_x} = 15.84 \text{ ms}$$

$$\begin{aligned} \text{Cycle time} &= \underbrace{(t_x + t_p)}_{\text{Frame}} + \text{Processing Delay} + \underbrace{(t_{xA} + t_p)} \\ &= (15.84 + 0.75) + 0.25 + (0.16 + 0.75) \text{ ms} \\ &= 17.75 \text{ ms} \end{aligned}$$

For Stop-and-Wait ARQ:

$$\begin{aligned} \text{Efficiency } (n) &= \frac{\text{Transmission delay}}{\text{Cycle Time}} * 100 \% \\ &= \frac{15.84 \text{ ms}}{17.75 \text{ ms}} * 100 \% = \boxed{89.23 \%} \end{aligned}$$

$$\underline{t_x} = \boxed{16 \text{ ms}}$$

$$\begin{aligned} \text{Cycle time} &= (t_x + t_p) + \text{Processing Delay} + (t_{xA} + t_p) \\ &= (16 + 0.75) + 0.25 + (0.16 + 0.75) \text{ ms} \\ &= 17.91 \text{ ms} \end{aligned}$$

For Stop-and-Wait ARQ:

$$\begin{aligned} \text{Efficiency } (\eta) &= \frac{\text{Transmission delay}}{\text{Cycle Time}} \times 100\% \\ &= \frac{16 \text{ ms}}{17.91 \text{ ms}} \times 100\% = \boxed{89.33\%} \end{aligned}$$

[MCQ]

[GATE-2023][1 Mark]



#Q. Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

- ☒ A Longer link length and lower transmission rate
- ☐ B Longer link length and higher transmission rate
- ☒ C Shorter link length and lower transmission rate
- ☒ D Shorter link length and higher transmission rate

Ans: B



stop-and-wait:-

$$\eta = \frac{t_x}{\text{cycle time}}$$

$$\eta = \frac{t_x}{(t_x + 2t_p)}$$

$$\eta = \frac{1}{1 + 2 * \left(\frac{t_p}{t_x}\right)}$$

$$\boxed{\eta = \frac{1}{1 + 2a}}$$

$a = \text{normalized propagation Delay} = \left(\frac{t_p}{t_x}\right)$

$$\eta = \frac{1}{1 + 2 * \left(\frac{t_p}{t_x}\right)} \quad \left| \begin{array}{l} \eta \propto t_x \\ \eta \propto \frac{1}{t_p} \end{array} \right.$$

$$\eta = \frac{1}{1 + 2 * \left(\frac{\text{Distance}}{\text{Signal speed}}\right) * \left(\frac{\text{Bandwidth}}{\text{Packet Size}}\right)}$$

$$\begin{array}{l} \eta \propto \text{Packet Size} \\ \eta \propto \text{signal speed} \end{array}$$

$$\eta \propto \frac{1}{\text{Bandwidth}}$$

$$\eta \propto \frac{1}{\text{Distance}}$$

#Q. A channel has a bit rate of 4 kbps and one-way propagation delay of 20 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of at least 50%, the minimum frame size should be

[GATE 2005]

☒ A 80 bytes

☒ B 80 bits

☒ C 160 bytes

☒ D 160 bits

Ans: D

Solution:-

$$\text{Bandwidth} = 4 \text{ Kbps} = 4 * 10^3 \text{ bits / sec}$$

$$t_p = 20 \text{ ms} = 20 * 10^{-3} \text{ Sec}$$

To achieve 50% utilization in Stop-and-Wait ARQ.

$$\text{Cycle time} = 2 * t_x$$

$$(t_x + 2 * t_p) = 2 * t_x$$

$$t_x = 2 * t_p$$

$$\text{Frame Size} = (2 * t_p) * \text{Bandwidth}$$

$$= (2 * 20 * 10^{-3} \text{ Sec}) * (4 * 10^3 \text{ bits / sec})$$

$$= 160 \text{ bits}$$

$$= 20 \text{ bytes}$$

#Q. Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgment and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is _____.

[GATE 2015, Set-1, 2-Mark]

IIT-K

IIT-K Ans: 160

Ans = 320

Solution:-

$$\text{Bandwidth} = 64 \text{ Kbps} = 64 * 10^3 \text{ bits / sec}$$

$$t_p = 20 \text{ ms} = 20 * 10^{-3} \text{ Sec}$$

To achieve 50% utilization in **Stop-and-Wait ARQ**.

$$\text{Cycle time} = 2 * t_x$$

$$(t_x + 2 * t_p) = 2 * t_x$$

$$t_x = 2 * t_p$$

$$\text{Frame Size} = (2 * t_p) * \text{Bandwidth}$$

$$= (2 * 20 * 10^{-3} \text{ Sec}) * (64 * 10^3 \text{ bits / sec})$$

$$= 2560 \text{ bits}$$

$$= 320 \text{ bytes}$$

#Q. A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay, and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one-way propagation delay (in milliseconds) is _____.

[GATE 2015, Set-2, 2-Mark]

Ans = 12

Solution:-

$$\text{Packet Size} = 1000 \text{ bytes} = 8 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{8 * 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 8 \text{ ms}$$

$$= 8 * 10^{-3} \text{ sec}$$

To achieve ^{25%}~~50%~~ utilization in Stop-and-Wait ARQ.

$$\text{Cycle time} = 4 * t_x$$

$$(t_x + 2 * t_p) = 4 * t_x$$

$$t_p = (3 * t_x / 2)$$

$$= (3 * 8 \text{ ms} / 2)$$

$$= 12 \text{ ms}$$

#Q. A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is _____ bytes/second.

[GATE 2016, Set-1, 2-Mark]

Ans = 2500

Solution:-

$$\text{Packet Size} = 1000 \text{ bytes} = 8 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 80 \text{ Kbps} = 8 * 10^4 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{8 * 10^3 \text{ bits}}{8 * 10^4 \text{ bits / sec}} = 100 \text{ ms}$$

$$t_p = 100 \text{ ms}$$

$$\text{ACK Size} = 100 \text{ bytes} = 8 * 10^2 \text{ bits}$$

$$\text{Bandwidth} = 8 \text{ Kbps} = 8 * 10^3 \text{ bits / sec}$$

$$t_{xA} = \frac{\text{ACK Size}}{\text{Bandwidth}} = \frac{8 * 10^2 \text{ bits}}{8 * 10^3 \text{ bits / sec}} = 100 \text{ ms}$$

$$\begin{aligned} \text{Cycle time} &= (t_x + t_p) + (t_{xA} + t_p) = 400 \text{ ms} \\ &= (100 + 100) + (100 + 100) \text{ ms} \end{aligned}$$

$$\text{Throughput} = \frac{\text{Packet Size}}{\text{Cycle Time}} = \frac{[1000 \text{ bytes}]}{[400 \text{ ms}]} = \frac{10 \text{ bytes}}{4 * 10^{-3} \text{ sec}}$$

$$= \boxed{2500 \text{ bytes / sec}} \checkmark$$

$$\eta = \frac{t_x}{\text{cycle time}} = \frac{100 \text{ ms}}{400 \text{ ms}} = \frac{1}{4}$$

$$\text{Throughput} = \eta * \text{Bandwidth}(\text{sender})$$

$$= \frac{1}{4} * 80 \text{ Kbps} = 20 \text{ Kbps} = 20 * 10^3 \text{ bits/sec} = \left(\frac{20 * 10^3}{8} \right) \frac{\text{bytes}}{\text{per sec}}$$

#Q. On a wireless link, the probability of packet error is 0.2. A stop-and-wait protocol is used to transfer data across the link. The channel condition is assumed to be independent from transmission to transmission. What is the average number of transmission attempts required to transfer 100 packets?

A 100

B 125

C 150

D 200

$$[100 + 20 + 4 + \dots] = 124$$

[GATE 2006]

$$\begin{aligned} \text{No. of packets } (n) &= 100 \\ \text{Packet-error prob. } (p) &= 0.2 \end{aligned}$$

$$\text{Total no. of transmission attempts } (N) = ?$$

$$N = n + (n \times p) + (n \times p^2) + (n \times p^3) + \dots$$

$$N = n [1 + p + p^2 + p^3 + \dots]$$

$$N = \frac{n}{(1-p)}$$

$$N = \frac{100}{(1-0.2)} = 125$$

Ans: B



Topic : Sliding Window ARQ

- Transmitter's transmitting window size = N $[N > 1]$
- Transmitter's transmit N frames continuously without any ACK
- Overlapping, unlike Stop-and-Wait ARQ
[To increase utilization]

$$\text{Efficiency} = \frac{[\text{Window Size} * \text{Packet transmission time}]}{\text{Cycle Time (RTT)}}$$

$$\eta = \frac{N * t_x}{\text{Cycle time}}$$



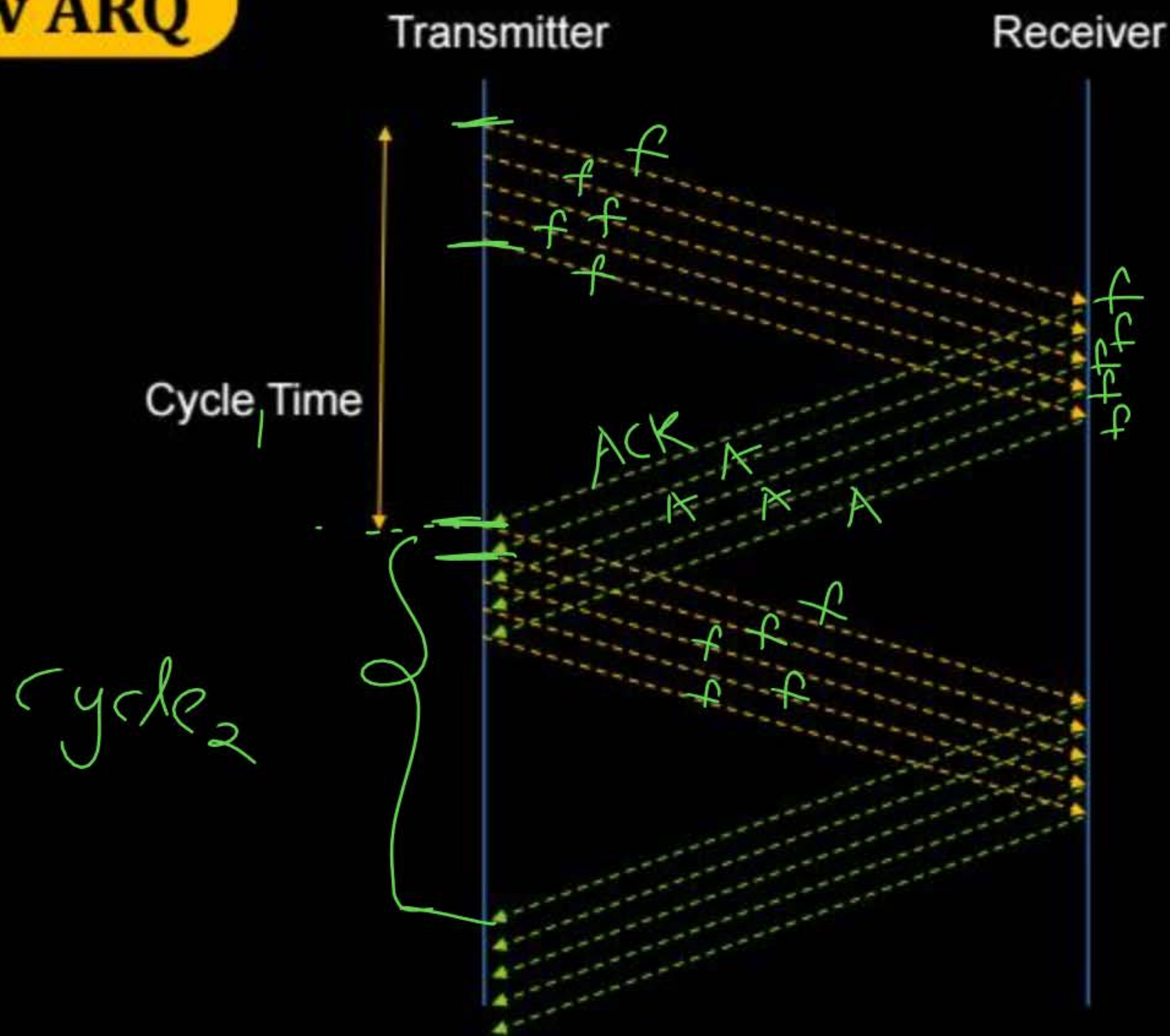
Topic : Sliding Window ARQ



$$\text{Efficiency}_{\text{Sliding Window}} = N * \text{Efficiency}_{\text{Stop-and-Wait}}$$



Topic : Sliding Window ARQ





Topic : Channel Utilization



- Link utilization or Throughput [in bits or bytes per sec]
- Total number of bytes (or bits) transmitted in Cycle time (RTT)

$$\text{Throughput} = \frac{[\text{Transmitter Window Size} * \text{Packet Size}]}{\text{Cycle Time (RTT)}}$$

$$\text{Throughput} = \text{Efficiency} * \text{Data transfer rate at transmitter}$$

#Q. Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is 50 microsecond. Acknowledgement packets (sent only from B to A) are very small and require negligible transmission time. The propagation delay over the link is 200 microsecond. What is the maximum achievable throughput in this communication?

[GATE 2003]

IIT-M
H.W.

- (A) $7.69 * 10^6$ bytes per sec
- (B) $11.11 * 10^6$ bytes per sec
- (C) $12.33 * 10^6$ bytes per sec
- (D) $15.00 * 10^6$ bytes per sec



Topic : Optimal Window Size



→ Optimal Window Size =

For maximum channel utilization,
minimum transmitter's transmitting window size.

$$\text{Optimal Window Size} = \left[\frac{\text{Cycle Time (RTT)}}{\text{Frame transmission time}} \right] \checkmark$$

$$\text{Optimal Window Size} = \frac{1}{\text{Efficiency}_{\text{[Stop-and-Wait]}}}$$

#Q. Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?

[GATE 2006]

IIT-KGP

A 20

B 40

C 160

D 320

Ans: B

Solution:-

$$\text{Packet Size} = 32 \text{ bytes} = 2^8 \text{ bits} = 2^5 \times 2^3 \text{ bits}$$

$$\text{Bandwidth} = 128 \text{ Kbps} = 2^7 * 10^3 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{2^8 \text{ bits}}{2^7 * 10^3 \text{ bits / sec}} = 2 \text{ ms}$$

$$\text{RTT (Cycle Time)} = 80 \text{ ms}$$

For Sliding Window ARQ :

$$\text{Optimal Window Size} = \left[\frac{\text{Cycle Time (RTT)}}{\text{Transmission delay}} \right]$$

$$= \left[\frac{80 \text{ ms}}{2 \text{ ms}} \right]$$

$$= 40$$



Topic : Sliding Window ARQ



→ Transmitter's transmitting window size = N $[N > 1]$

→ Receiver's receiving window size = N

→ Total number of sequences = N $[0 \text{ to } (N-1)]$

Total number of sequences
= Transmitter's transmitting window size

Sequence number $\leftarrow$ (Frame number) mod (N)



2 mins Summary



Topic

Sliding Window Protocol



THANK - YOU

